

DX Enabler를 위한

오렌지를 활용한 데이터 분석 과정

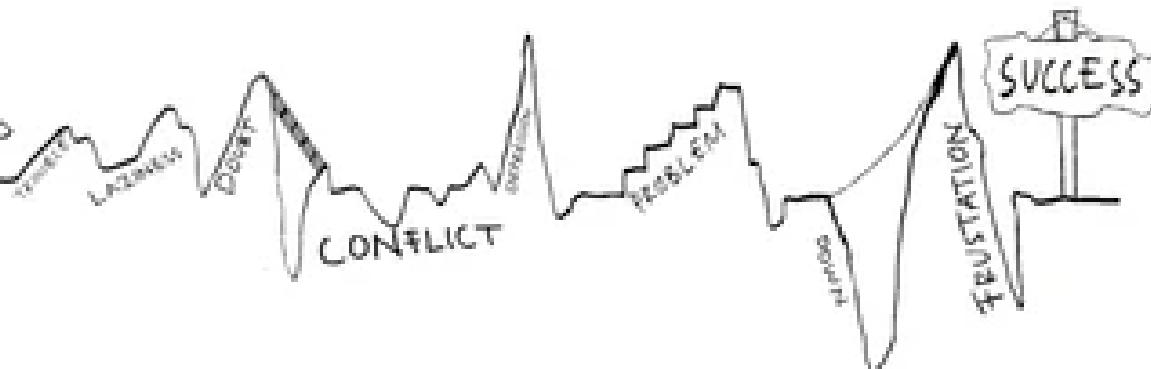


여러분은 어떤 목적을 갖고 이번 과정에 참석하였습니까?

-EXPECTATION-



-REALITY-





1. DX 과제를 제안한다.
2. 자신의 업무를 DX로 개선한다.
3. 전략 과제의 팀원으로 역량을 발휘한다.
4. DX 과제를 리딩한다.
5. 끊임없이 학습한다.





지식

“우리가 알고 이해하는 것 ”

- 전통적 지식 (예, 수학, 물리...)
- 학제 간 연구
- 현대적 지식 (예, 디지털 기술)
- themes (예, Digital Literacy)



스킬

“아는 것을 사용하는 방법 ”

- 창의력 Creative Thinking
- 비판적 사고 Critical Thinking
- 의사 소통 Communication
- 협업 Collaboration



인성

“행동하고 세계에 참여하는 방법 ”

- 마음챙김
- 호기심
- 용기
- 회복탄력성
- 윤리성
- 리더십

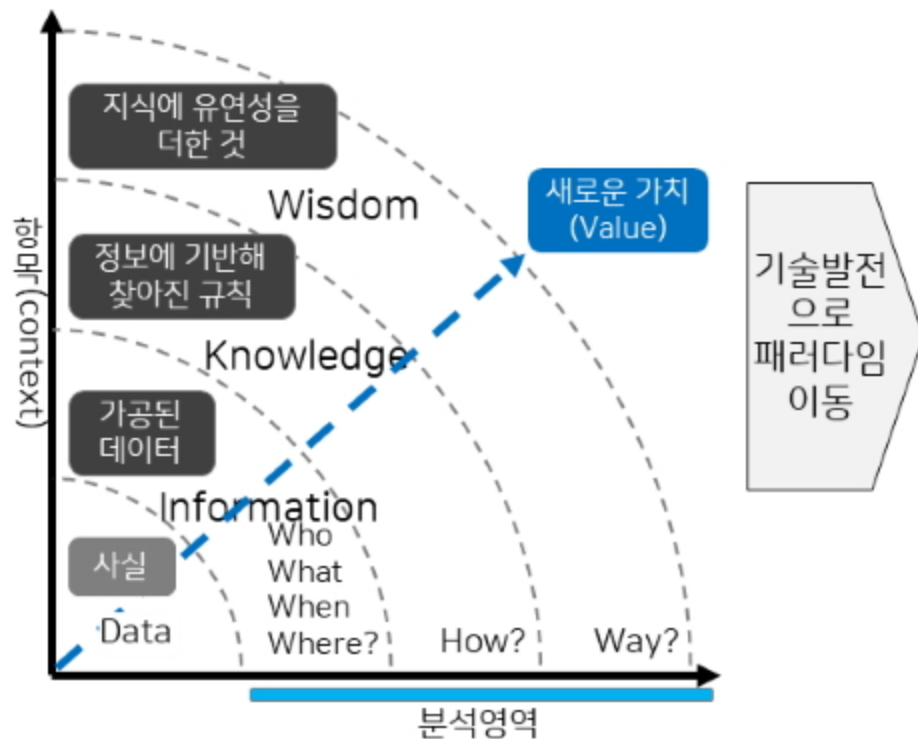
메타학습

“반영하고 학습하는 방법”, “학습하는 방법에 대한 학습 ”

- 메타 인지 자신이 아는지 모르는지를 성찰하는 힘
- 성장 마인드셋 노력이 재능이나 지능을 향상시킬 수 있다는 믿음

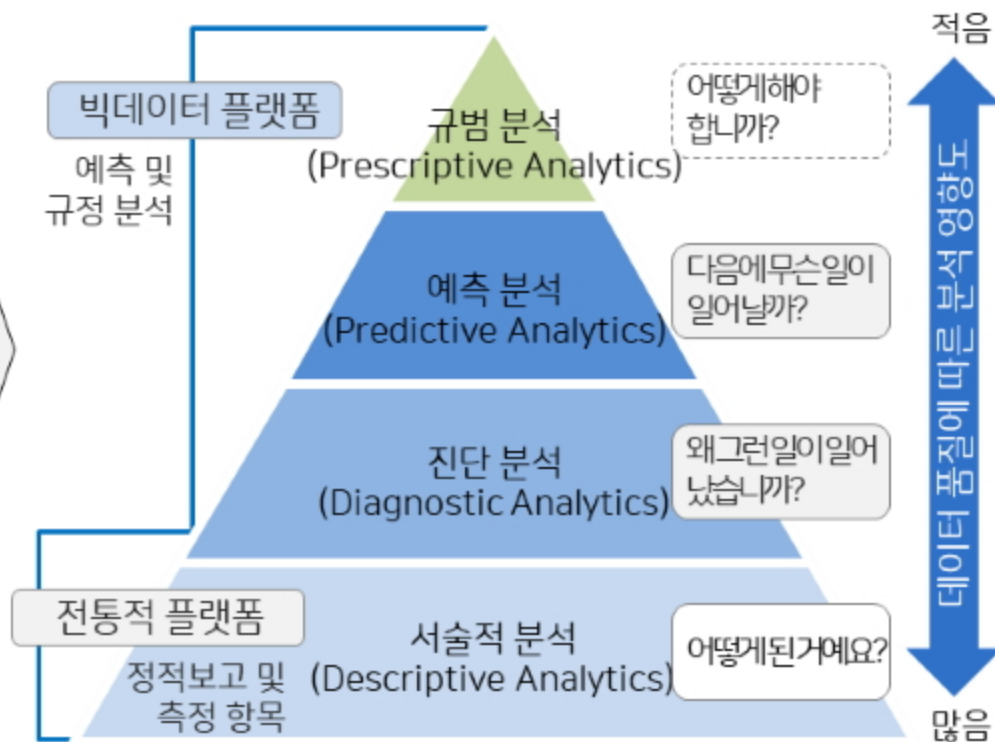


개념적 지식화 모델(DIKW 모델)



출처 : 로리 제니퍼 (2007). "지혜 계층 : DIKW 계층의 표현 " 자료 재구성

빅데이터 분석 모델



출처 : Gartner 자료 재구성



1. 기초적인 ML 활용 방법을 이해한다.

→ Learning by Doing, 데이터 분석 심화

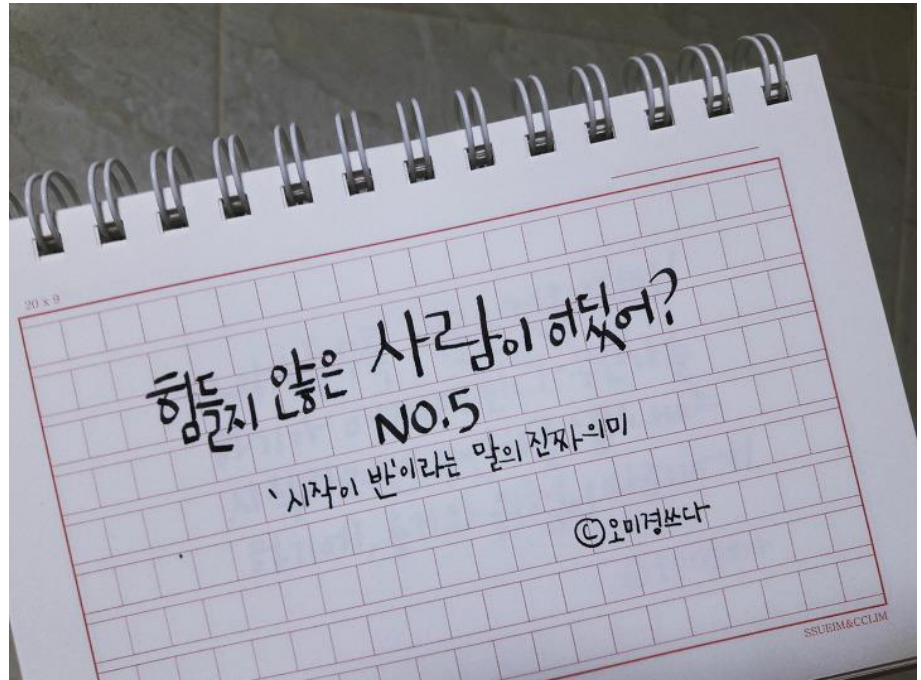
2. 표준적인 과제 실행 절차를 검토한다.

→ 과제 적합형 실행 절차 Tailoring 논의





1. introduction
2. Data 분석 관련 용어
3. Data Handling
4. 탐색적 분석과 데이터 시각화
5. Machine Learning(1) 분류분석
6. Machine Learning(2) 회귀분석
7. Machine Learning(3) 주성분분석
8. Machine Learning(4) 군집분석
9. Machine Learning(5) Image 분석
10. Machine Learning(6) Text 분석
11. [검토] DX 과제 실행 절차



IRIS 품종 분류

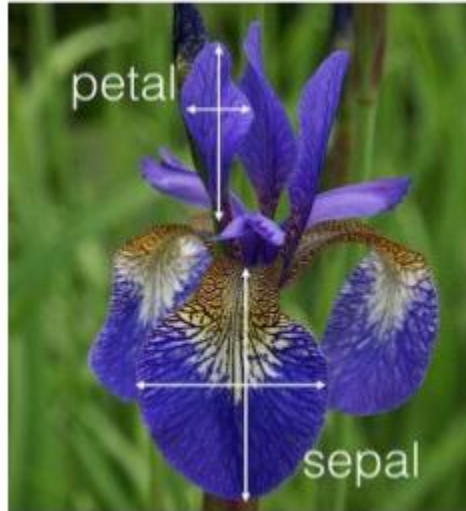
문제

sepal(꽃받침)의 폭과 길이
petal(꽃잎)의 폭과 길이를 가지고
품종(species)을 분류(예측)

문제 정의

Y : species (3종 : setosa, versicolor, virginica)

X : petal_length
petal_width
sepal_length
sepal_width



Training / test data

Features				Labels
Sepal length	Sepal width	Petal length	Petal width	Species
5.1	3.5	1.4	0.2	Iris setosa
4.9	3.0	1.4	0.2	Iris setosa
7.0	3.2	4.7	1.4	Iris versicolor
6.4	3.2	4.5	1.5	Iris versicolor
6.3	3.3	6.0	2.5	Iris virginica
5.8	3.3	6.0	2.5	Iris virginica

iris setosa



petal sepal

iris versicolor



petal sepal

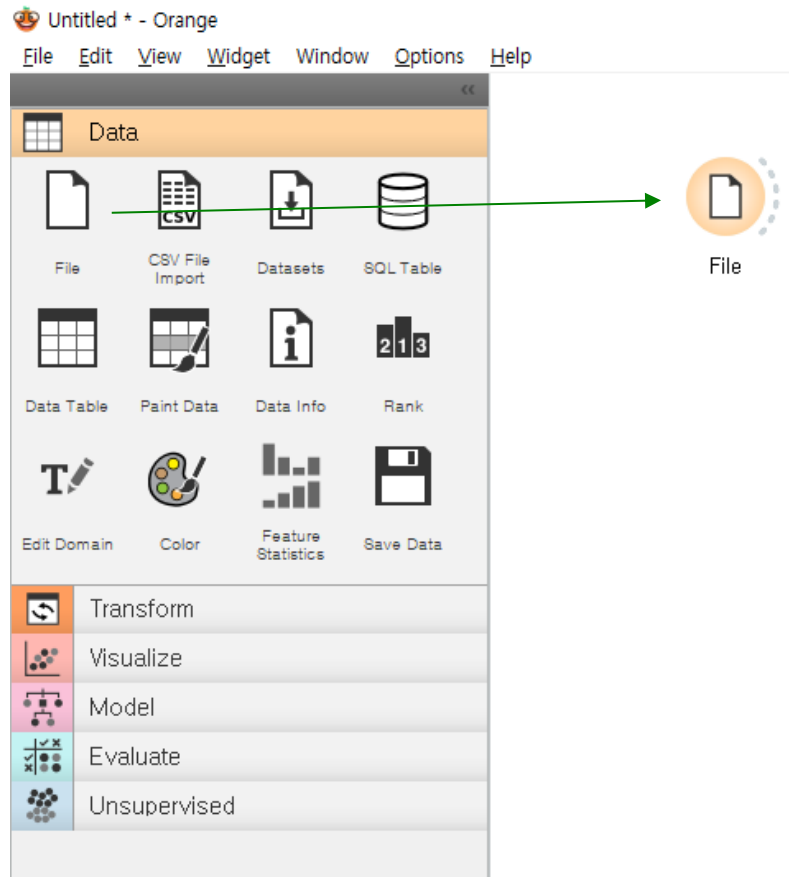
iris virginica



petal sepal

데이터를 불러오고, Data size, 변수 명, 변수 형태를 확인합니다.

실습 파일 : IRIS_전처리실습.csv



The screenshot shows the 'File' widget configuration window. The 'Source' is set to 'File: IRIS_전처리실습.csv'. The 'File Type' is set to 'Automatically detect type'. The 'Info' section shows '150 instances' and '5 features (2.0% missing values)'. The 'Columns' table is shown below.

① Data를 저장한 폴더를 찾아서 Data 명 더블 클릭

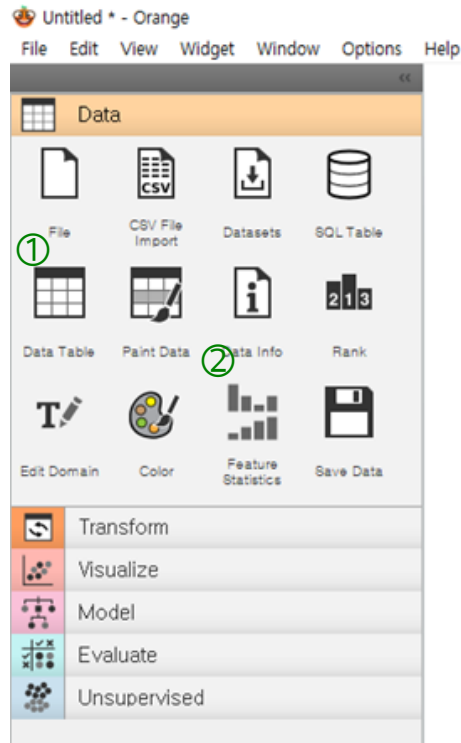
② Data 개요 확인

Name	Type	Role	Values
1 sepal_length	numeric	feature	
2 sepal_width	numeric	feature	
3 petal_length	numeric	feature	
4 petal_width	numeric	feature	
5 species	categorical	target	setosa, versicolor, virginica

③ 변수 유형 확인

④ 변수의 역할 확인
→ species를 target으로 변경 (target=Y, 결과변수, 종속변수)

데이터 전체를 살펴 보고, 기초 통계량을 확인합니다. 특히 Missing Value 확인



Data Table - Orange

File Edit View Window Help

Info
150 instances
4 features (2.0 % missing data)
Target with 3 values (2.0 % missing data)
No meta attributes.

Variables
☒ Show variable labels (if present)
☐ Visualize numeric values
☒ Color by instance classes

Selection
☒ Select full rows

	species	sepal_length	sepal_width	petal_length	petal_width
1	setosa	?	3.5	1.4	0.2
2	setosa	4.9	?	1.4	0.2
3	setosa	4.7	3.2	?	0.2
4	setosa	4.6	3.1	1.5	?
5	?	5.0	3.6	1.4	0.2
6	setosa	5.4	3.9	1.7	0.4
7	setosa	4.6	3.4	1.4	0.3
8	setosa	5.0	3.4	1.5	0.2
9	setosa	4.4	2.9	1.4	0.2
10	setosa	4.9	3.1	1.5	0.1
11	setosa	5.4	3.7	1.5	0.2
12	setosa	4.8	3.4	1.6	0.2
13	setosa	4.8	3.0	1.4	0.1
14	setosa	4.3	3.0	1.1	0.1
15	setosa	5.8	4.0	1.2	0.2
16	setosa	5.7	4.4	1.5	0.4
17	setosa	5.4	3.9	1.3	0.4
18	setosa	5.1	3.5	1.4	0.3
19	setosa	5.7	3.8	1.7	0.3
20	setosa	5.1	3.8	1.5	0.3
21	setosa	5.4	3.4	1.7	0.2
22	setosa	5.1	3.7	1.5	0.4

Feature Statistics - Orange

File View Window Help

각 변수에 missing value가 3개씩 있다.

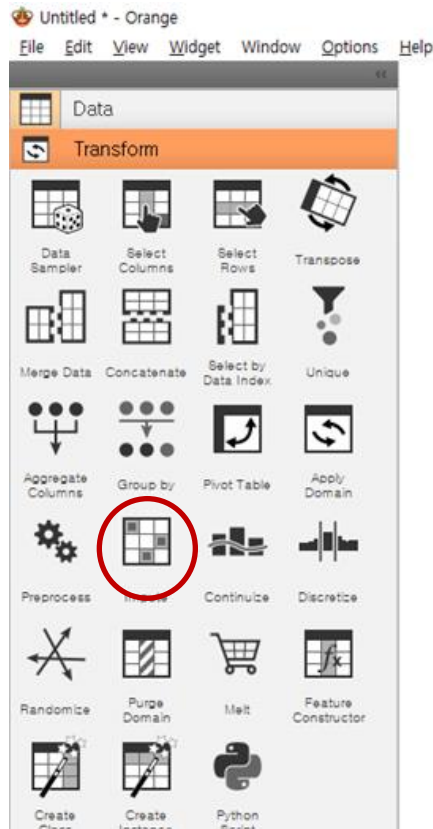
	Name	Distribution	Mean	Mode	Median	Dispersion	Min.	Max.	Missing
N	sepal_length		5.837	5.0	5.8	0.141	4.3	7.9	3 (2 %)
N	sepal_width		3.059	3.0	3.0	0.143	2.0	4.4	3 (2 %)
N	petal_length		3.752	1.4	4.3	0.468	1.0	6.9	3 (2 %)
N	petal_width		1.201	0.2	1.3	0.634	0.1	2.5	3 (2 %)
C	species			setosa		1.1			3 (2 %)

Color: None

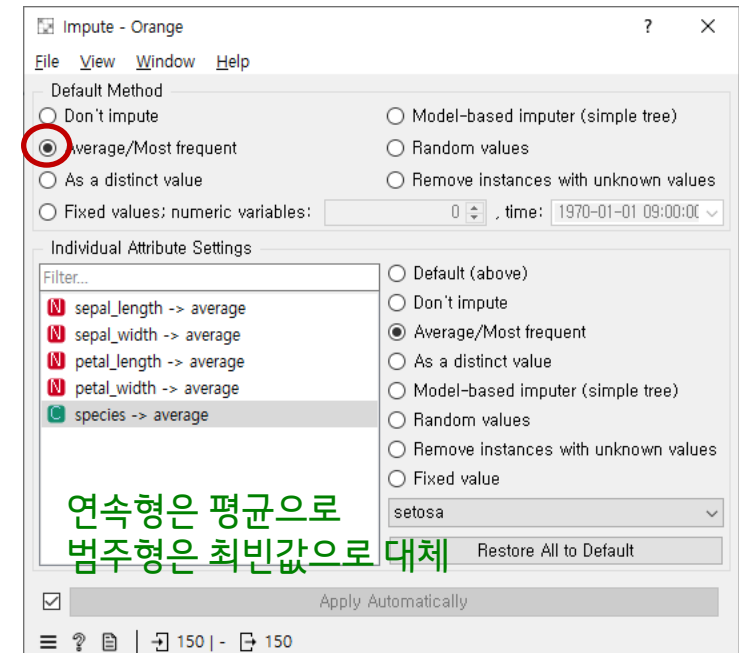
☒ Send Automatically

150 5

결측치를 다른 값으로 대체하고, 데이터를 확인하여 결측값이 처리되었음을 확인합니다.

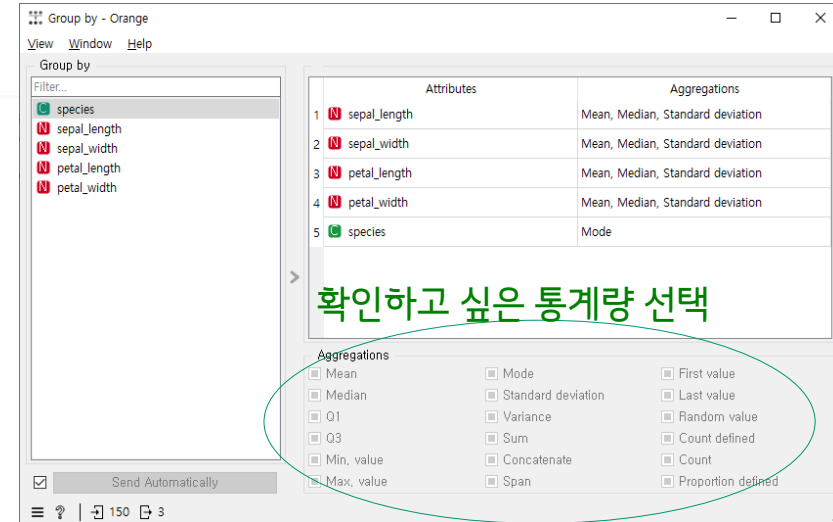
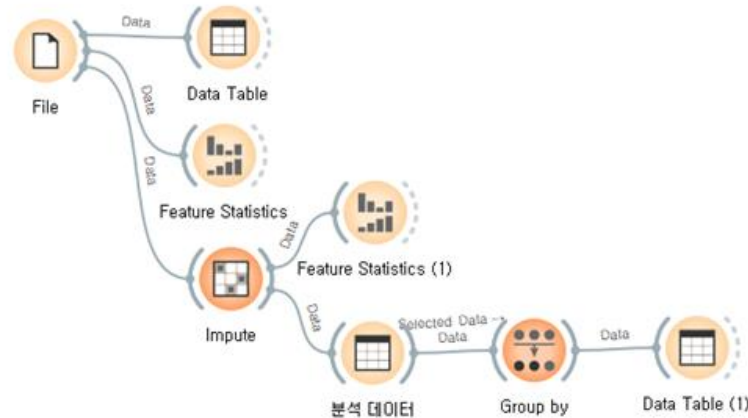
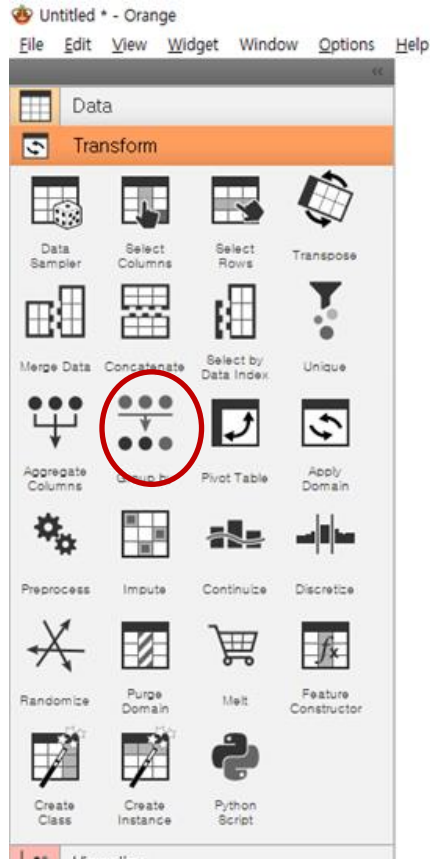


오른쪽 마우스로 rename 가능
※ 결함이 없는 데이터로 분석한다.



연속형은 평균으로
범주형은 최빈값으로 대체

품종 별 통계량을 확인합니다.



Data Table (1) - Orange

File Edit View Window Help

Info

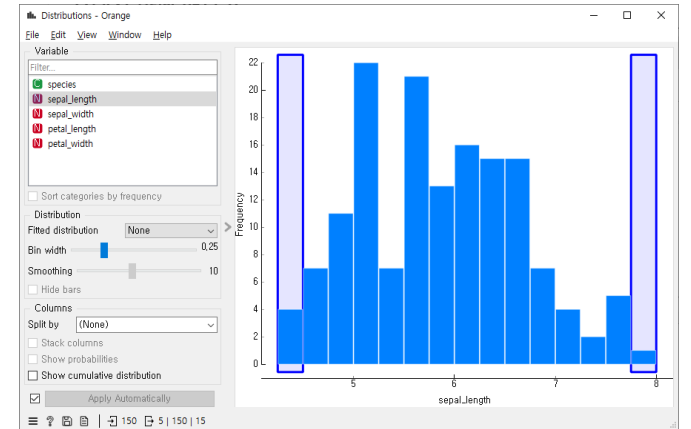
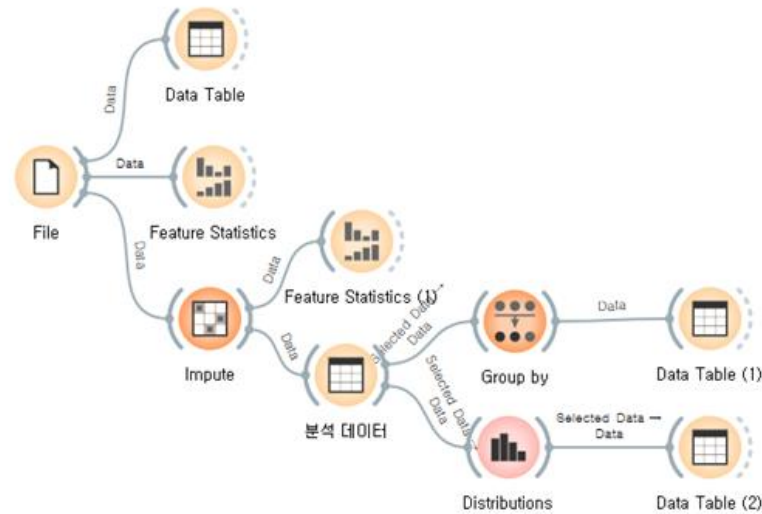
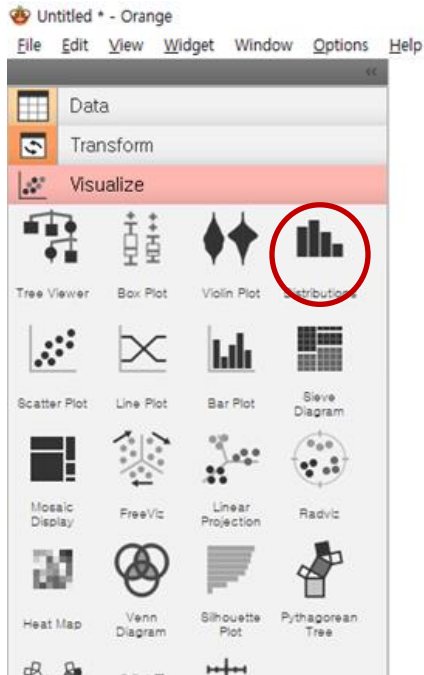
3 instances (no missing data)
12 features
No target variable.
2 meta attributes

Variables

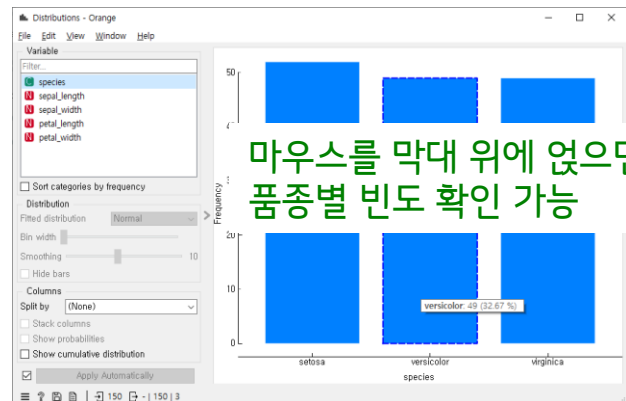
Show variable labels (if present)

	species - Mode	species	pal_length - Me	pal_length - Med	gth - Standard	pal.
1	setosa	setosa	5.07764	5	0.46374	
2	versicolor	versicolor	5.90076	5.83741	0.490618	
3	virginica	virginica	6.58036	6.5	0.650042	

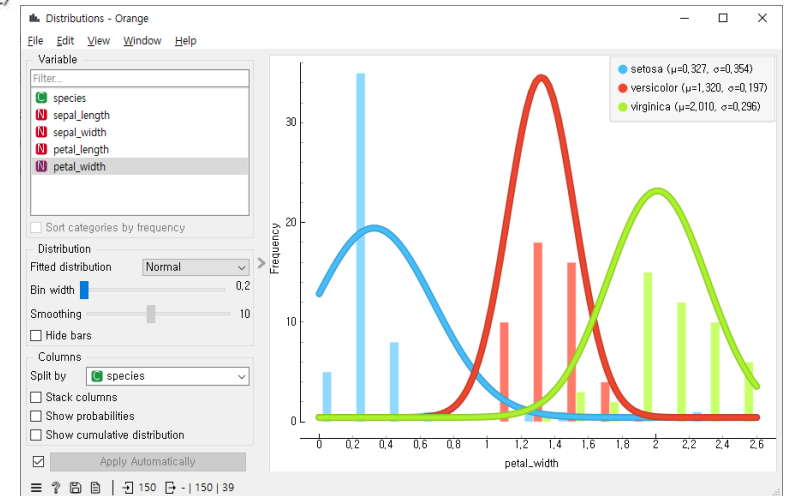
단변량의 분포를 확인합니다.



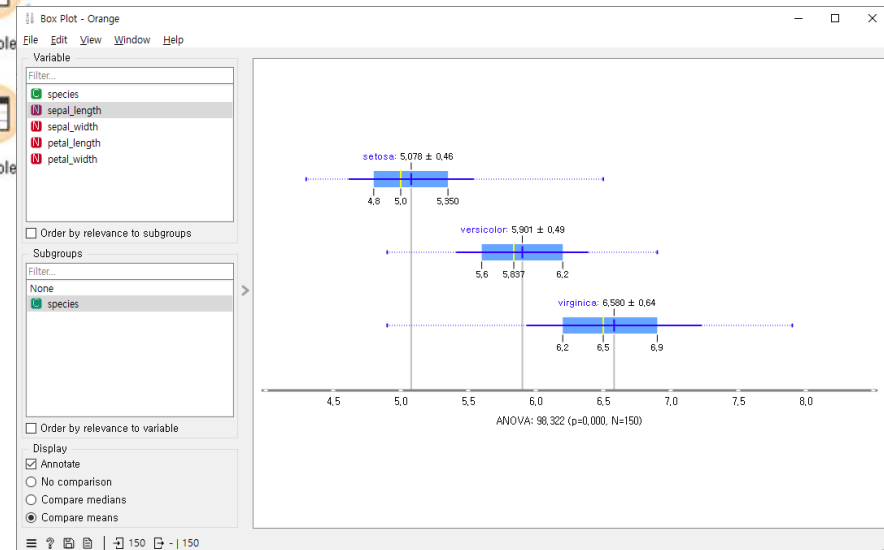
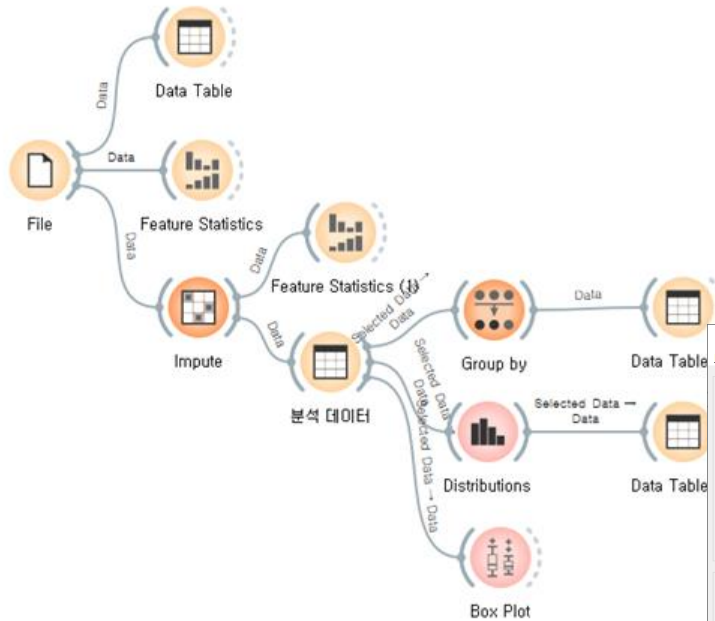
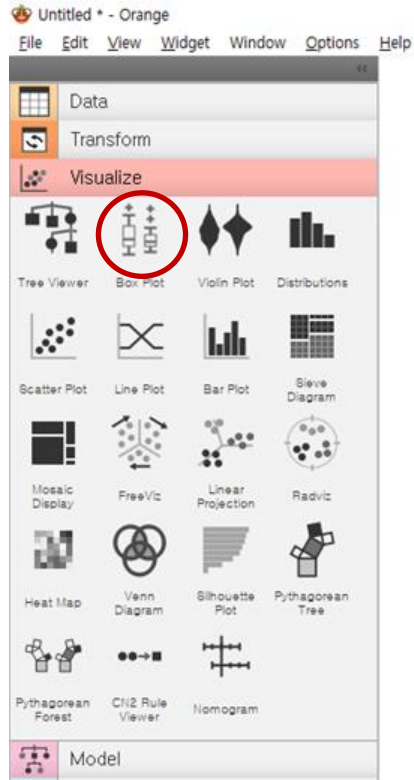
	species	Selected	sepal_length	sepal_width	petal_length	petal_width
1	setosa	sepal_length < ...	4.4	2.9	1.4	0.2
2	setosa	sepal_length < ...	4.3	3.0	1.1	0.1
3	setosa	sepal_length < ...	4.4	3.0	1.3	0.2
4	setosa	sepal_length < ...	4.4	3.2	1.3	0.2
5	virginica	sepal_length ≥ ...	7.9	3.8	6.4	2.0



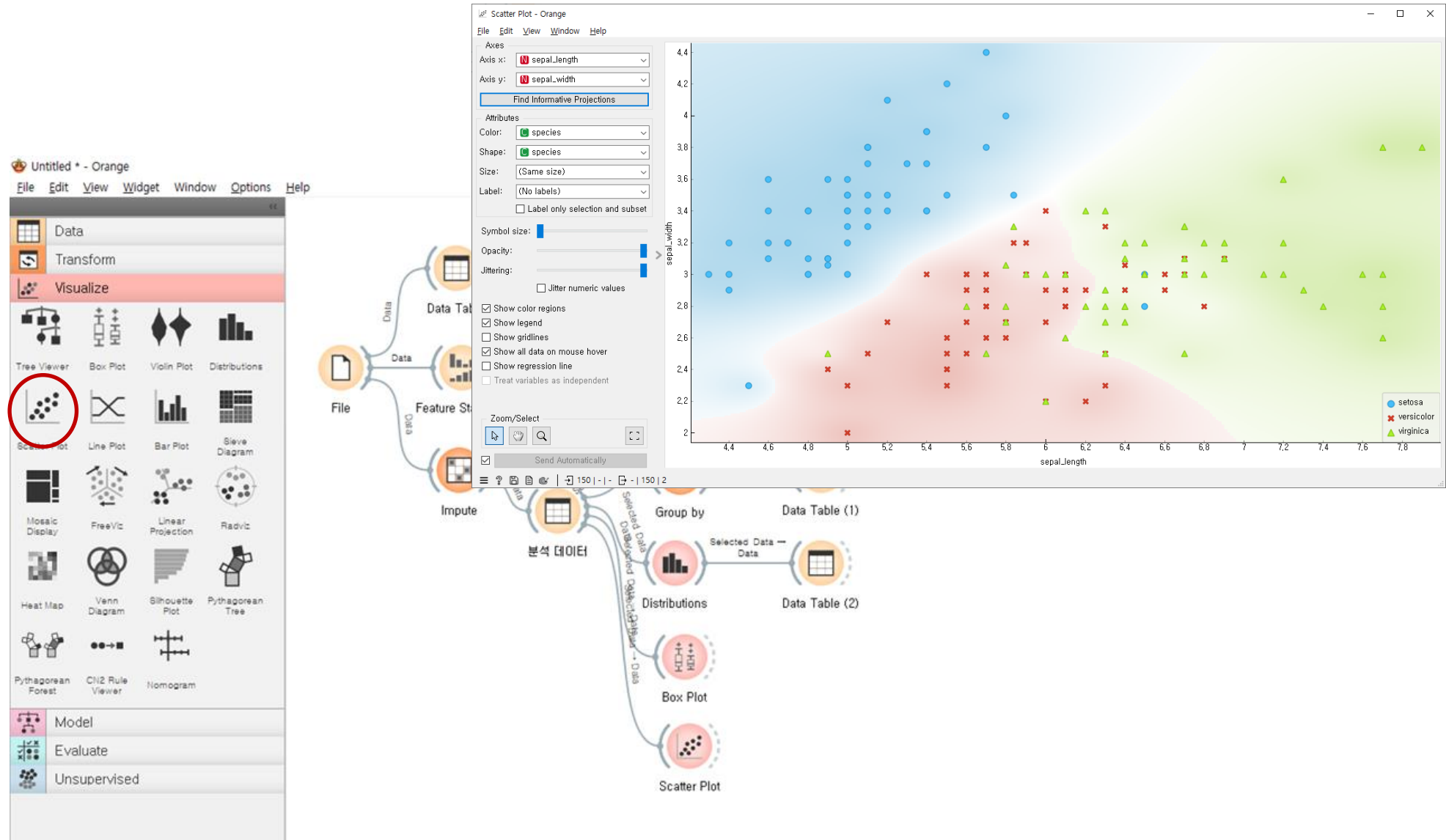
마우스를 막대 위에 얹으면
품종별 빈도 확인 가능



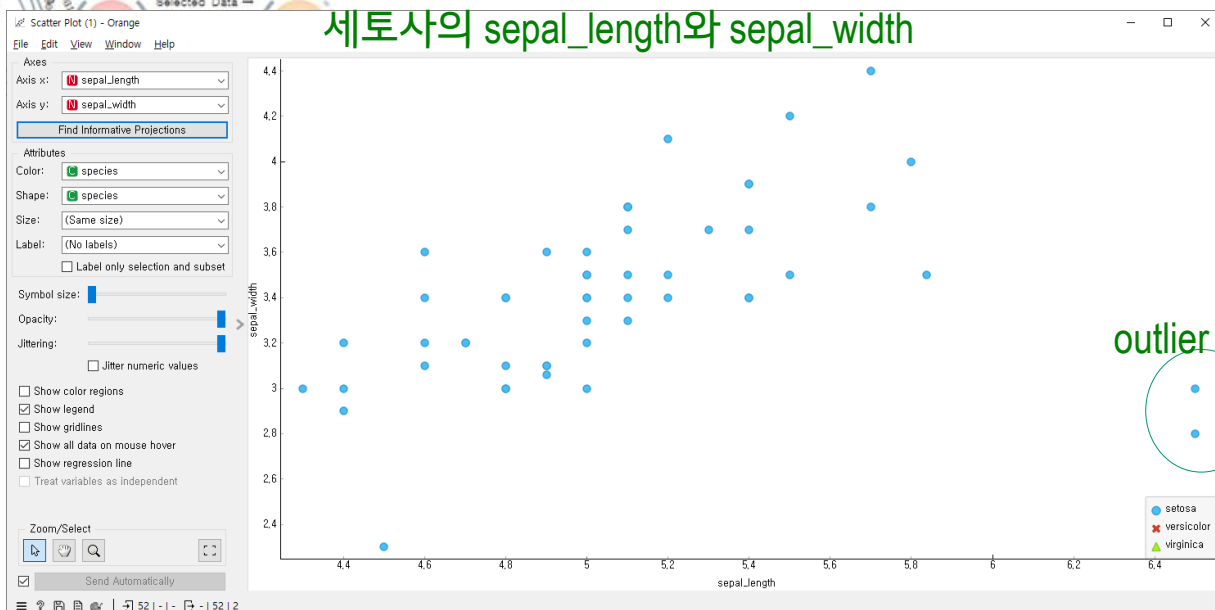
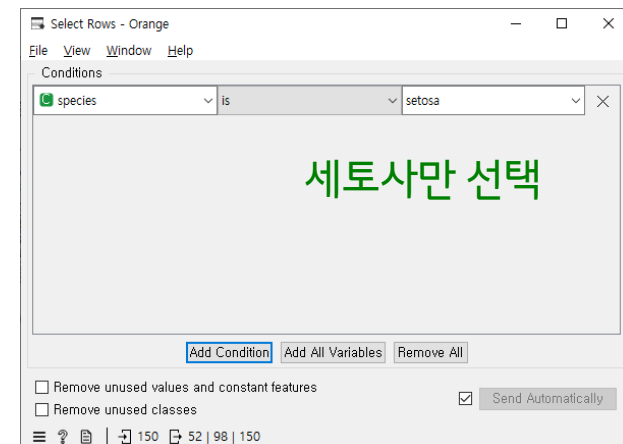
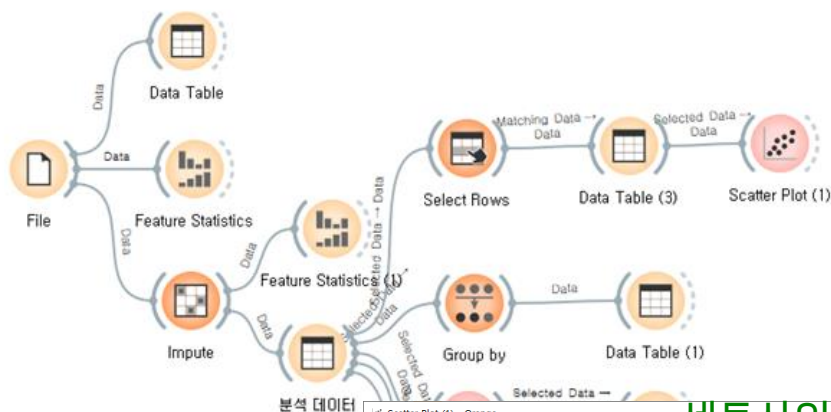
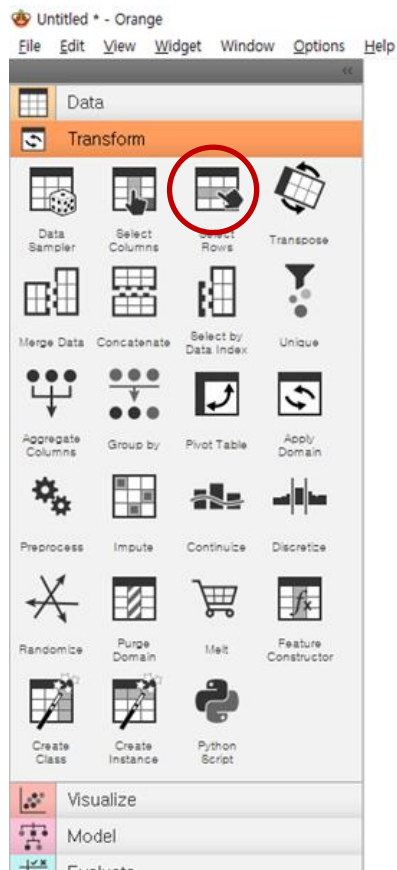
품종 별 각 특성을 비교합니다.



특성 간의 관계를 확인합니다.

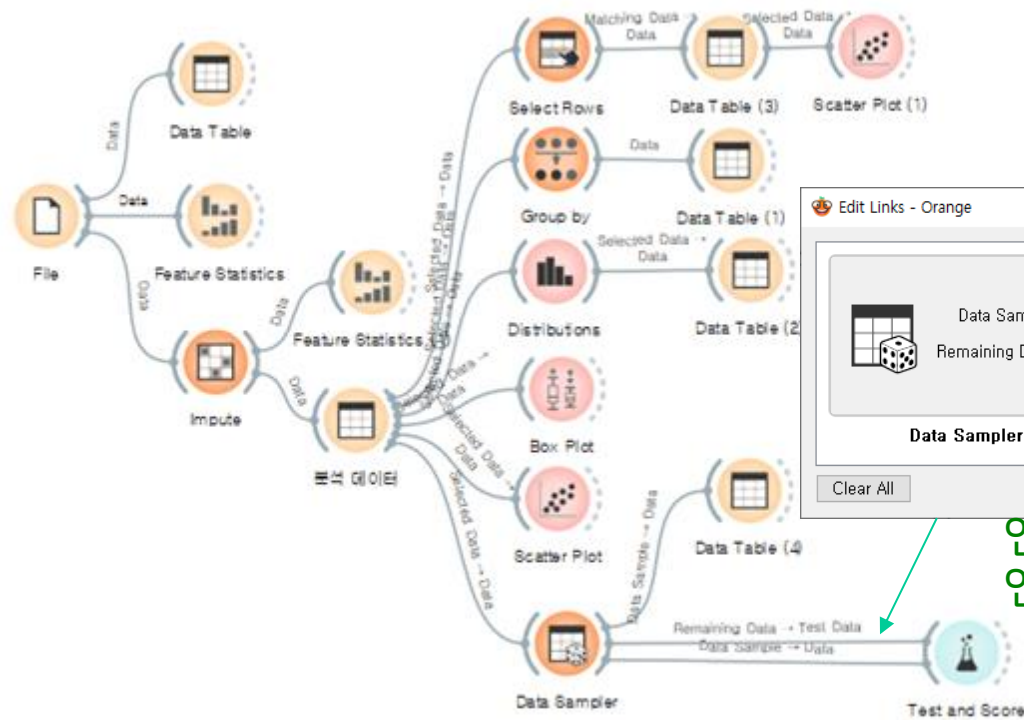


품종 별 관계를 탐색합니다.



분류 모형을 만들려면, 데이터를 훈련용과 검증용으로 나눕니다.

The screenshot shows the Orange3 software interface. At the top, the title bar reads "Untitled * - Orange". Below it is a menu bar with "File", "Edit", "View", "Widget", "Window", "Options", and "Help". The main area displays a list of widget categories on the left: "Data", "Transform", "Visualize", "Model", "Evaluate", and "Unsupervised". The "Evaluate" category is selected and highlighted in light blue. Within the "Evaluate" category, several widgets are listed: "Test and Score" (highlighted with a red circle), "Predictions", "Confusion Matrix", "ROC Analysis", "Performance Curve", and "Calibration Plot". Each widget has a small icon representing its function.



연결선을 클릭하여
연결해 준다.

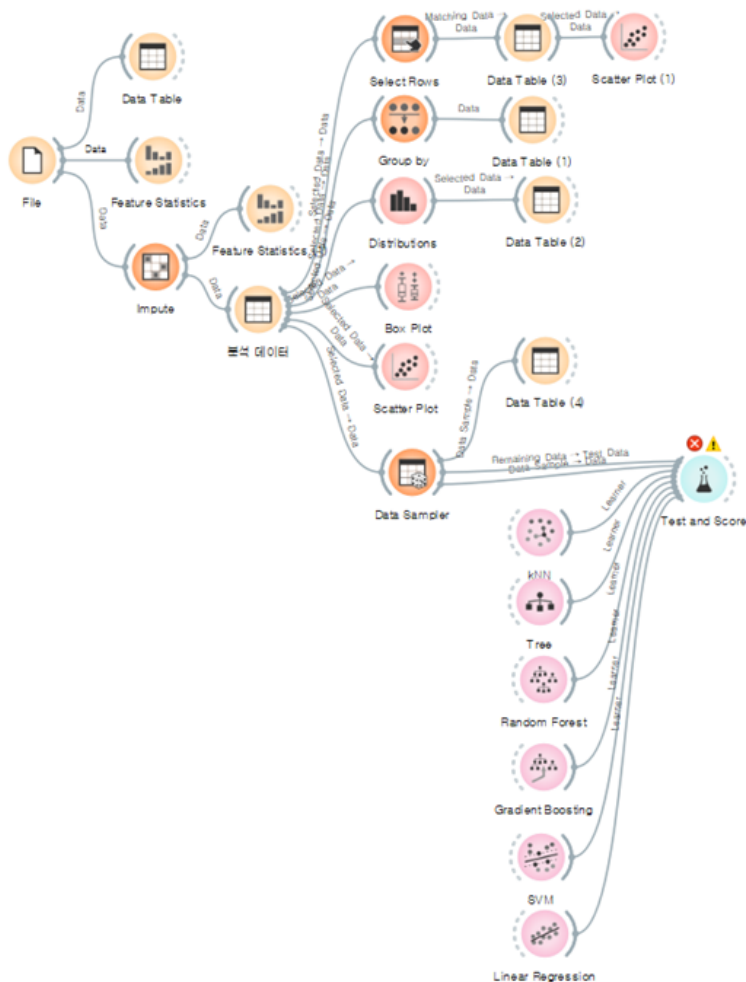
모형 개발을 위해 학습시킬 알고리즘들을 연결하고 비교 평가

Untitled * - Orange
File Edit View Widget Window Options Help

Widget palette:

- Data: Data Table
- Transform: Feature Statistics, Impute
- Visualize: Distributions, Box Plot, Scatter Plot
- Model:
 - Constant, CH2 Rule Induction, Calibrated Learner, kNN
 - Tree, Random Forest, Gradient Boosting, SVM
 - Linear Regression, Logistic Regression, Naive Bayes, AdaBoost
 - Curve Fit, Neural Network, Stochastic Gradient Descent, Stacking
- Save Model, Load Model
- Evaluate: Unsupervised

Test and Score
Cross-validation accuracy estimation.



Test and Score - Orange

File Edit View Window Help

☐ Cross validation
 Number of folds: 5
☒ Stratified
☐ Cross validation by feature
☐ Random sampling
 Repeat train/test: 10
 Training set size: 66 %
☒ Stratified
☐ Leave one out
☐ Test on train data
☒ Test on test data

Evaluation results for target (None, show average over classes)

Model	AUC	CA	F1	Prec	Recall	MCC
kNN	0.924	0.911	0.913	0.929	0.911	0.871
Tree	0.914	0.844	0.847	0.892	0.844	0.786
Random Forest	0.978	0.911	0.913	0.929	0.911	0.871
Gradient Boosting	0.944	0.889	0.889	0.904	0.889	0.838
SVM	0.980	0.933	0.934	0.944	0.933	0.903
Linear Regression						

Compare models by: Area under ROC curve ☐ Negligible diff.: 0.1

	kNN	Tree	Random Fo...	Gradient Bo...	SVM
kNN					
Tree					
Random Forest					
Gradient Boosting					
SVM					

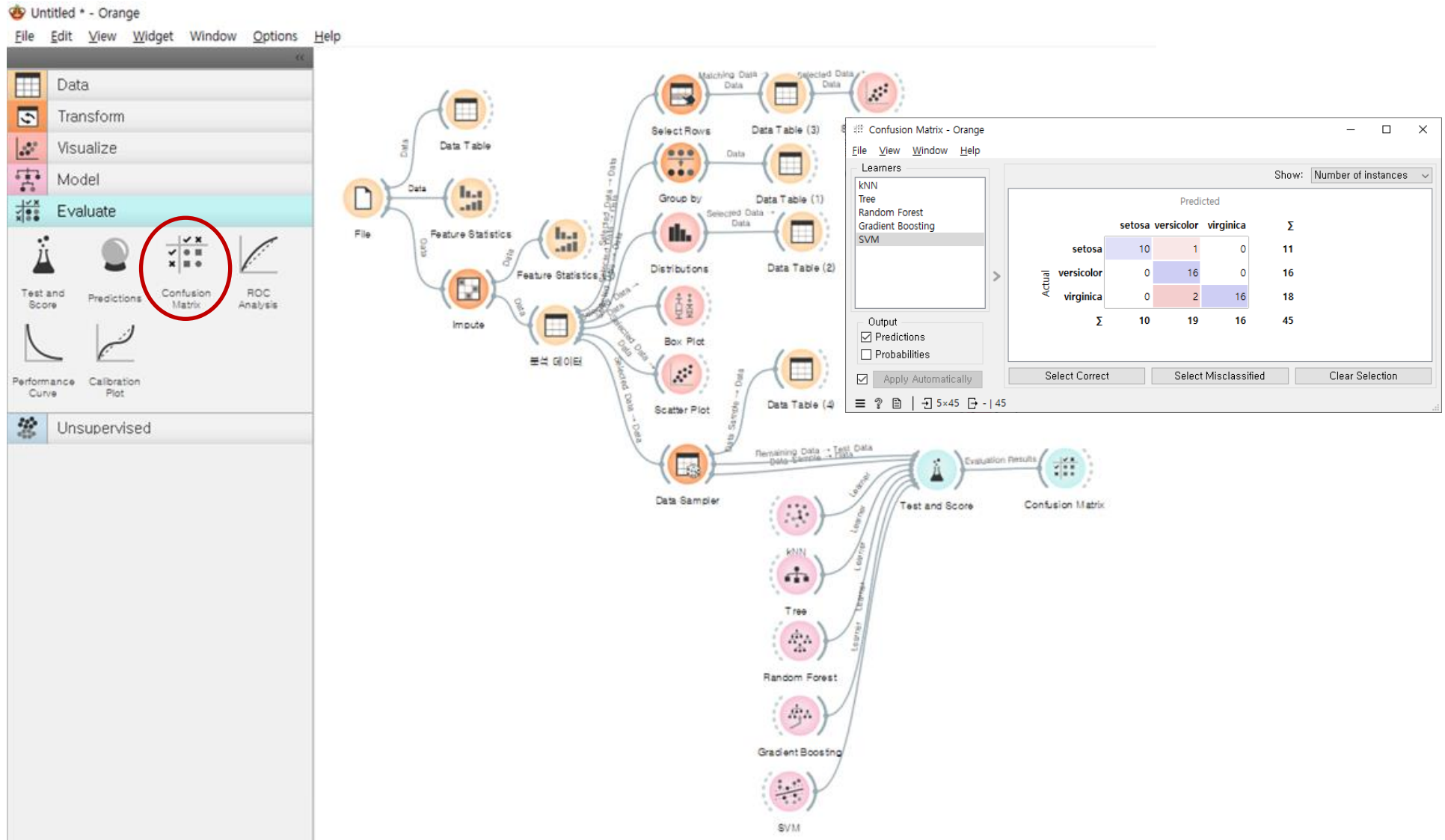
Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

Linear Regression failed with error: (1 warning)

SVM이 가장 높은 값.
즉, 가장 좋은 모형

linear regression은 에러.
분류분석에 사용할 수 없는
알고리즘

Confusion Matrix를 통해 오분류의 내용을 확인합니다.



예측하고 싶은 데이터를 가져오고, Predictions 위젯을 연결

Untitled * - Orange

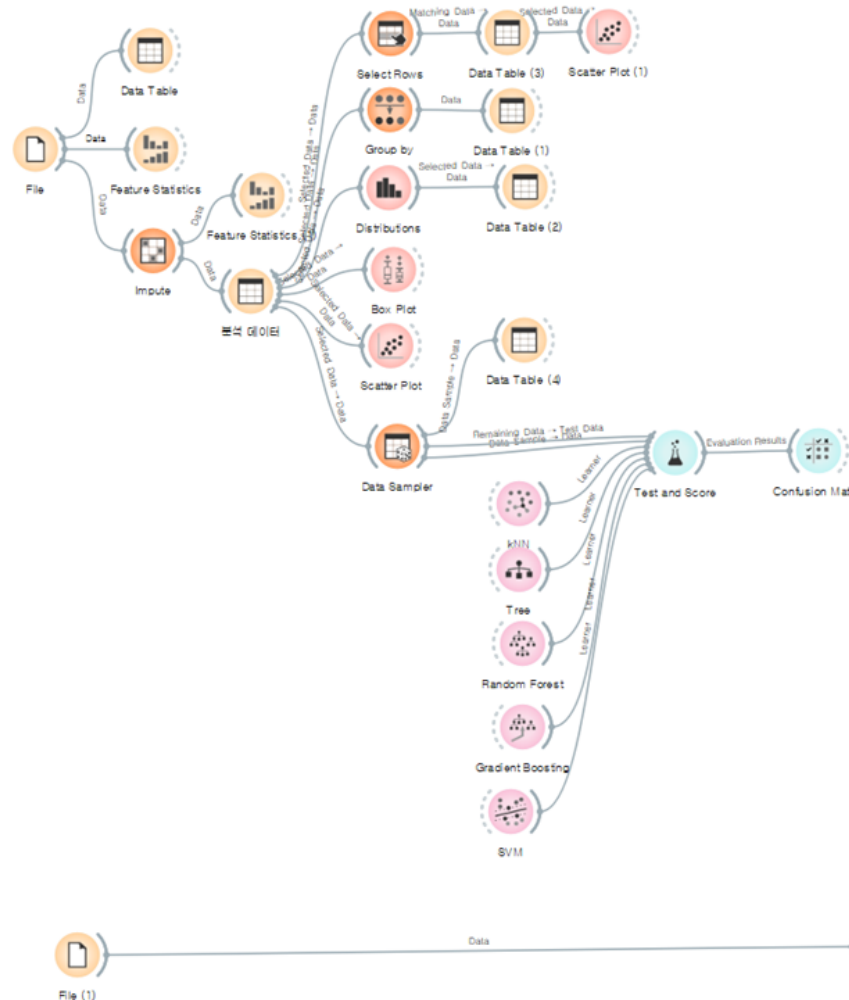
File Edit View Widget Window Options Help

Data
Transform
Visualize
Model
Evaluate

Test and Score
Predictions
Confusion Matrix
ROC Analysis
Performance Curve
Calibration Plot

Unsupervised

Select a widget to show its description.
See [workflow examples](#), [YouTube tutorials](#), or open the [welcome screen](#).



데이터 파일 : IRIS예측

File (1) - Orange

File View Window Help

Source
● File: IRIS예측.xlsx ... Reload

○ URL:

File Type
Automatically detect type

Info
3 instances
4 features (no missing values)
Data has no target variable,
0 meta attributes

Columns (Double click to edit)

Name	Type	Role	Values
1 sepal_length	numeric	feature	
2 sepal_width	numeric	feature	
3 petal_length	numeric	feature	
4 petal_width	numeric	feature	

꽃잎과 꽃받침 데이터로
품종을 예측하고자 한다.

Rese... Apply

Browse documentation datasets

개발된 모델을 연결해주고, 예측 결과를 확인합니다.

Untitled - Orange

File Edit View Widget Window Options Help

Data
Transform
Visualize
Model
Evaluate

Test and Score
Predictions
Confusion Matrix
ROC Analysis
Performance Curve
Calibration Plot
Unsupervised

Select a widget to show its description.
See [workflow examples](#), [YouTube tutorials](#), or open the [welcome screen](#).

The workflow diagram shows the following steps:

- Data** widget connects to **Data Table**.
- Data Table** connects to **Feature Statistics**.
- Feature Statistics** connects to **Impute**.
- Impute** connects to **Feature Statistics (2)**.
- Feature Statistics (2)** connects to **Matching Data**, **Select Rows**, **Data Table (3)**, **Scatter Plot (1)**, **Group by**, **Data Table (1)**, **Distributions**, **Data Table (2)**, **Box Plot**, **Scatter Plot**, **Data Table (4)**, **Data Sampler**, and **SVM**.
- Data Sampler** connects to **Test and Score**.
- SVM** connects to **Test and Score**.
- Test and Score** connects to **Predictions**.

Predictions - Orange

File View Window Help

Show probabilities for (None) [Restore Original Order](#)

	sepal_length	sepal_width	petal_length	petal_width
1 versicolor	6.4	2.8	1.9	2.1
2 virginica	2.5	3.0	2.3	3.3
3 setosa	4.7	3.2	1.3	0.2

품종 예측 결과

File (1) Predictions



Question

Comment

Suggestion